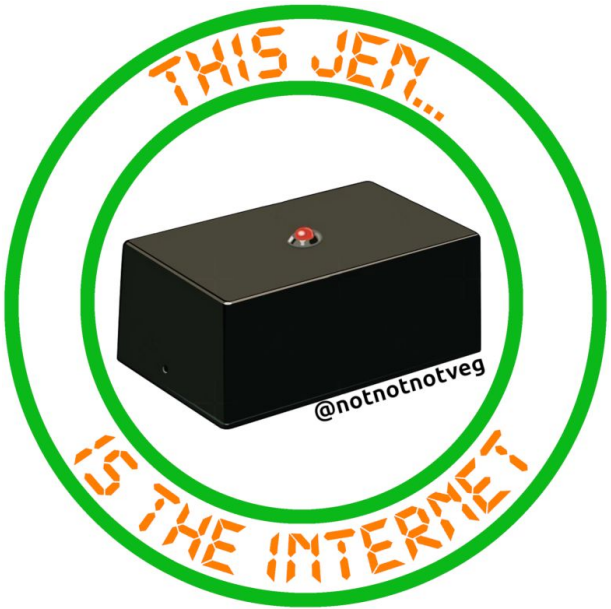
Browser-Based Port Scanning in the Age of LNA

Gev Manekshaw | BSidesNYC 2026

WhoAmI

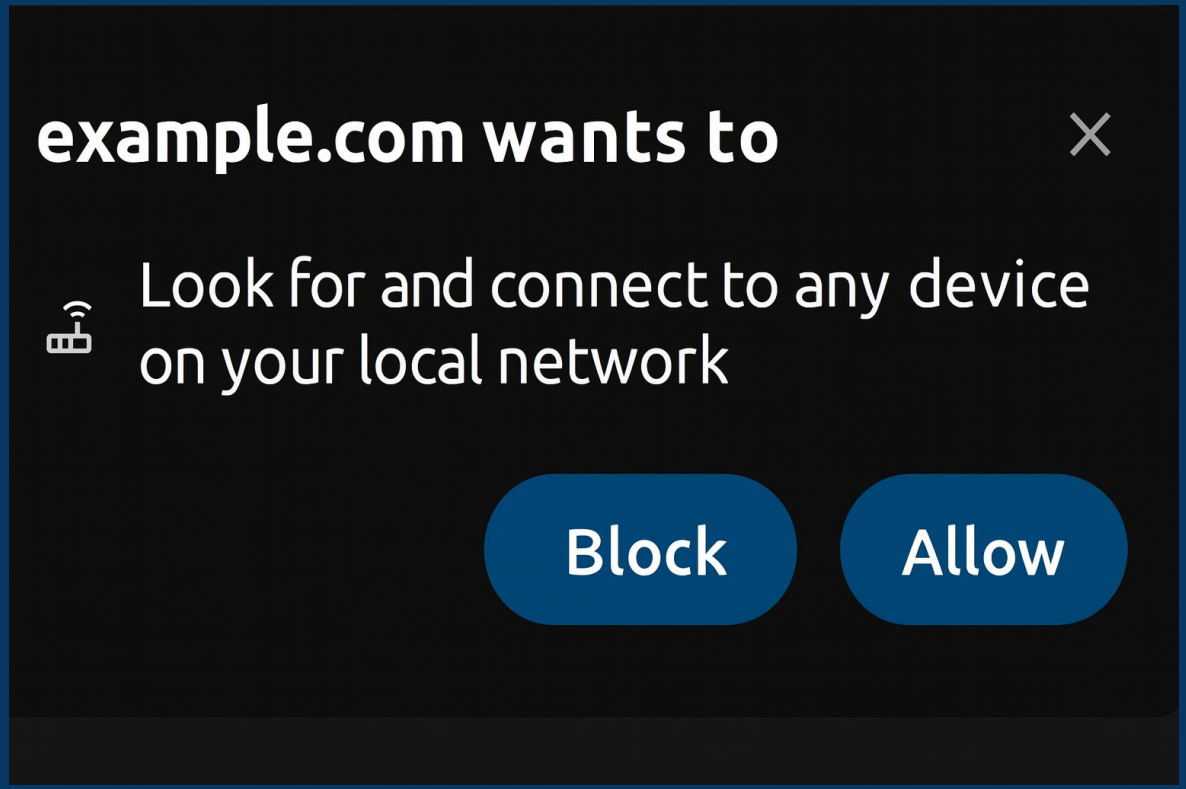
Gev Manekshaw

- Security Engineer & Researcher
- Philly based
- BSides Philly Speaker
- 30+ Hall of Fames for VDPs

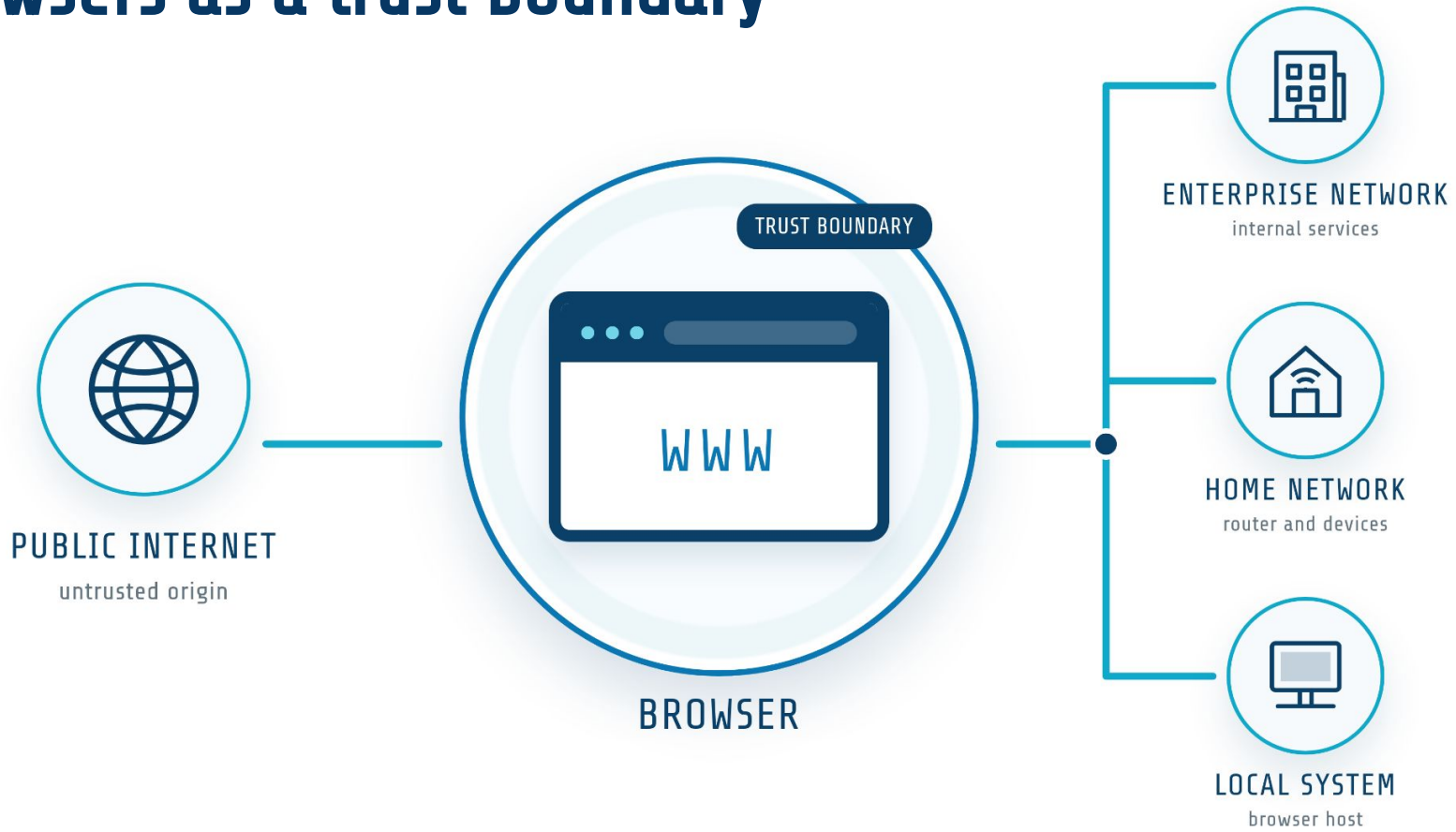


Disclaimer :

- Views expressed in this talk do not represent those of my employer.
- Tools shared are intended to be used in your own or trusted environments.


example.com wants to
Look for and connect to any device on your local network
Block
Allow

Browsers as a trust boundary



Acting: “Drive-by” attacks

SOHO pharming :

```
fetch(“https://router.local/setDNS?server=1.2.3.4")
```

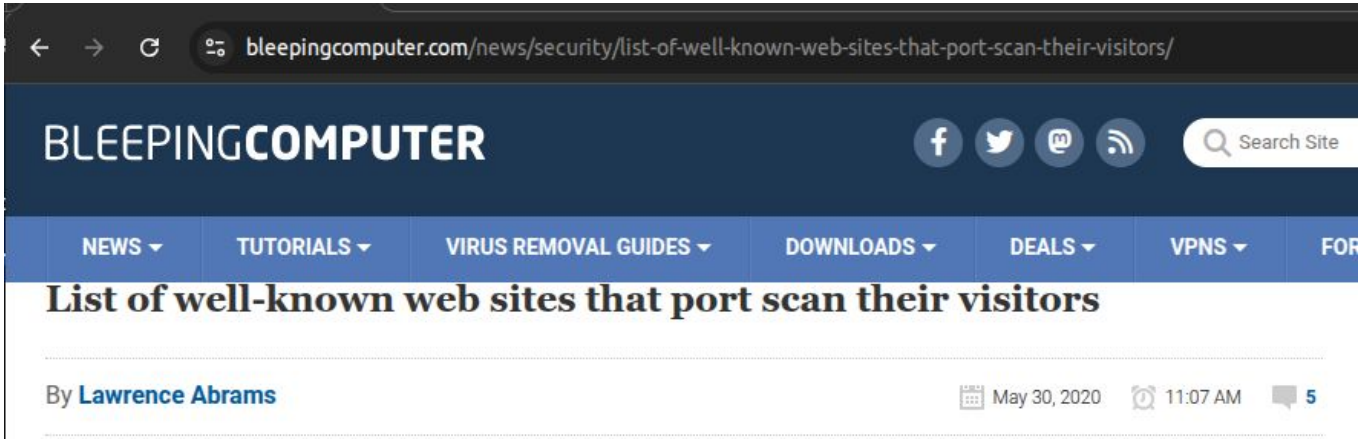
CVE-2025-49596 :

```
fetch(“http://mcp-insp.local:6277/sse?command=cat&args=%2Fetc%2Fpasswd")
```

Observing: Port Scanning

eBay Community > Buying & Selling > Selling > Dear Ebay, Why are you scanning my ports?

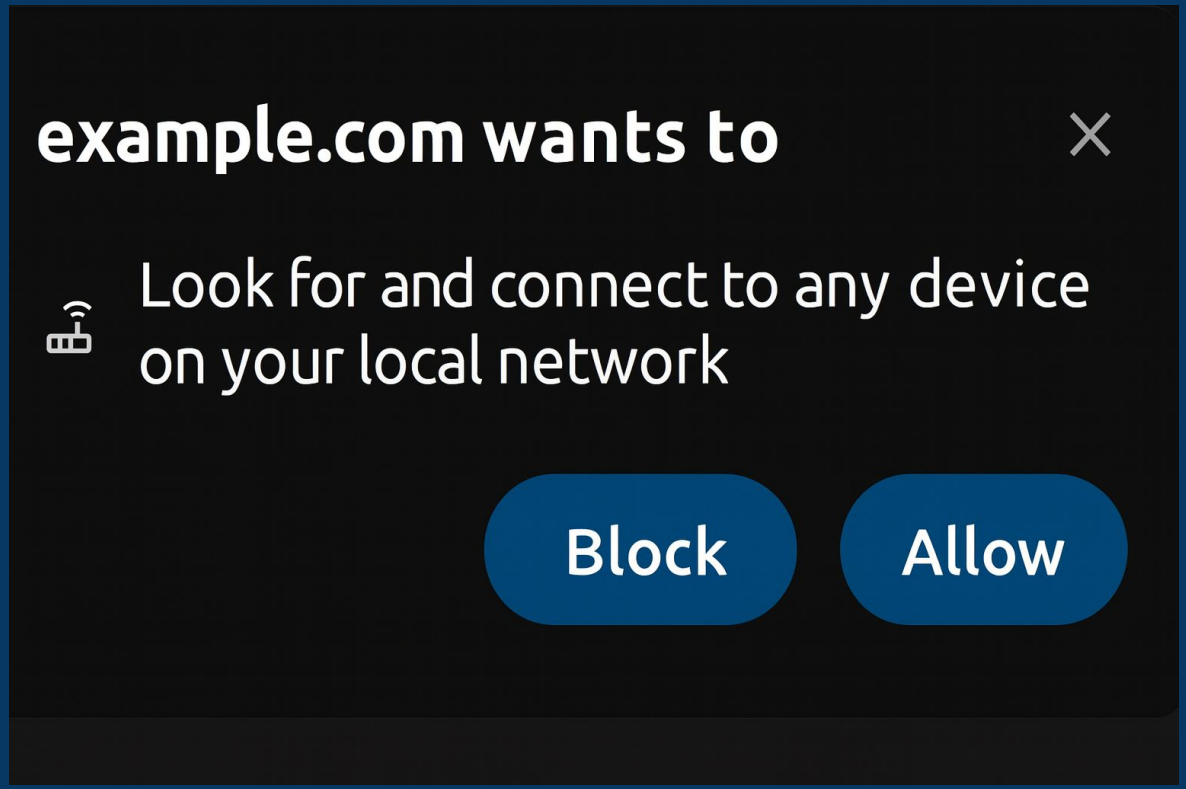
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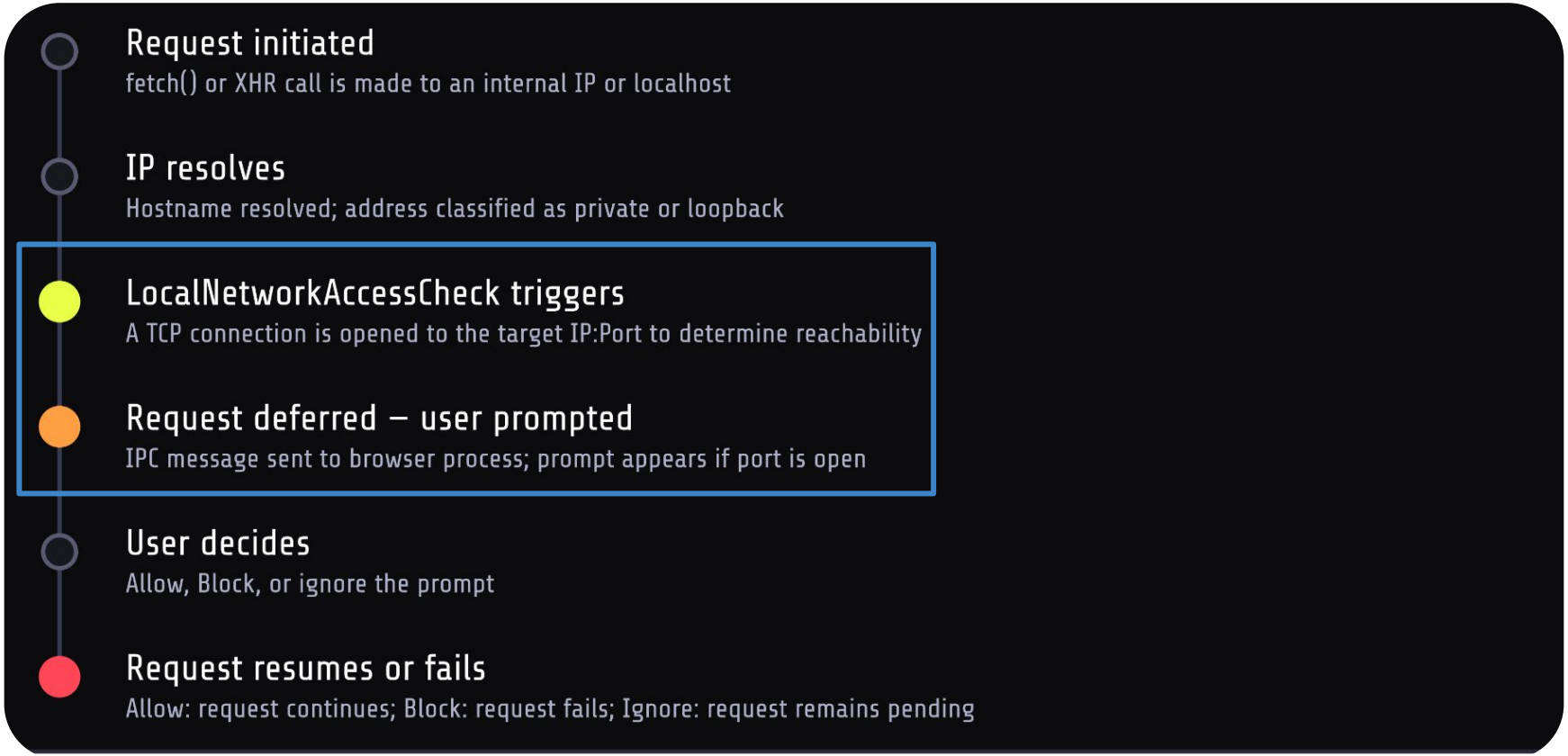
Local Network Access (LNA)

Public websites need *user permission* before reaching localhost or a private network.

Introduced in Oct 2025 (Chrome 142)


example.com wants to
Look for and connect to any device on your local network
Block
Allow

The LNA Prompt Path



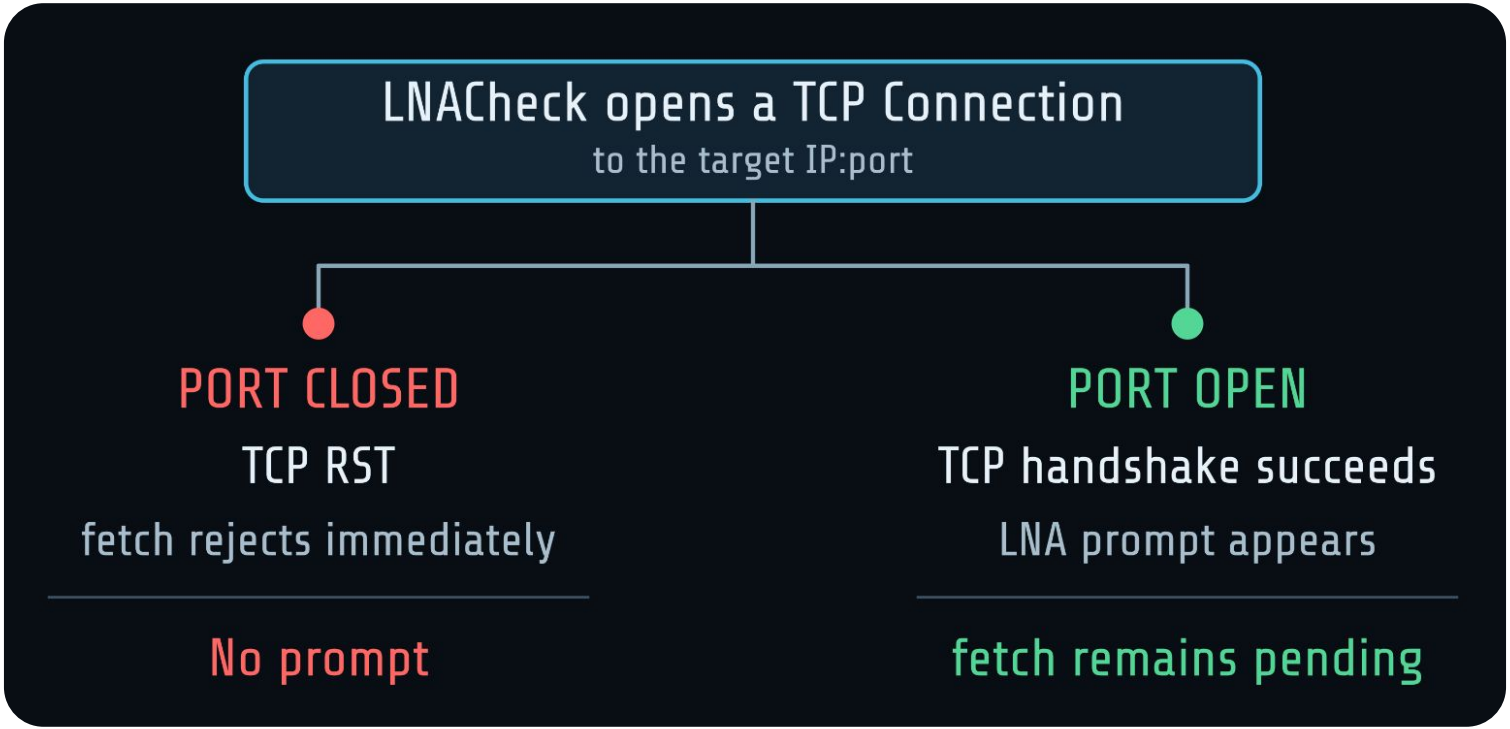
The browser has to know*

whether a port is reachable

before deciding whether to prompt.

** In current implementations*

LNACheck: The Port Probe

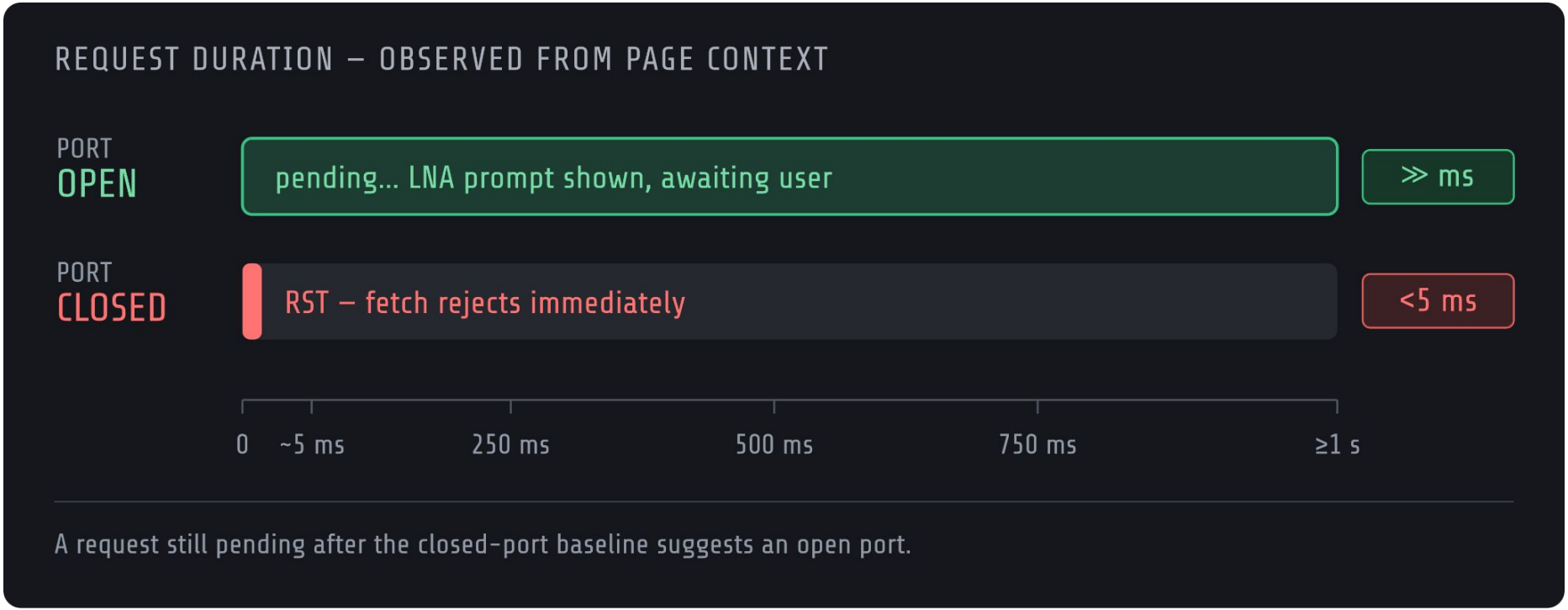


The webpage's JS can infer
which path the request takes.



Time

The Timing Split



State of LNA Prompt vs. Port Scannability

LNA state	Port can be probed
Allowed	Yes
Blocked	No
Pending Selection	Yes

State of LNA Prompt vs. Port Scannability

LNA state	Port can be probed
Allowed	Yes
Blocked	No
Pending Selection	Yes ¯_(ツ)_/¯

PoC

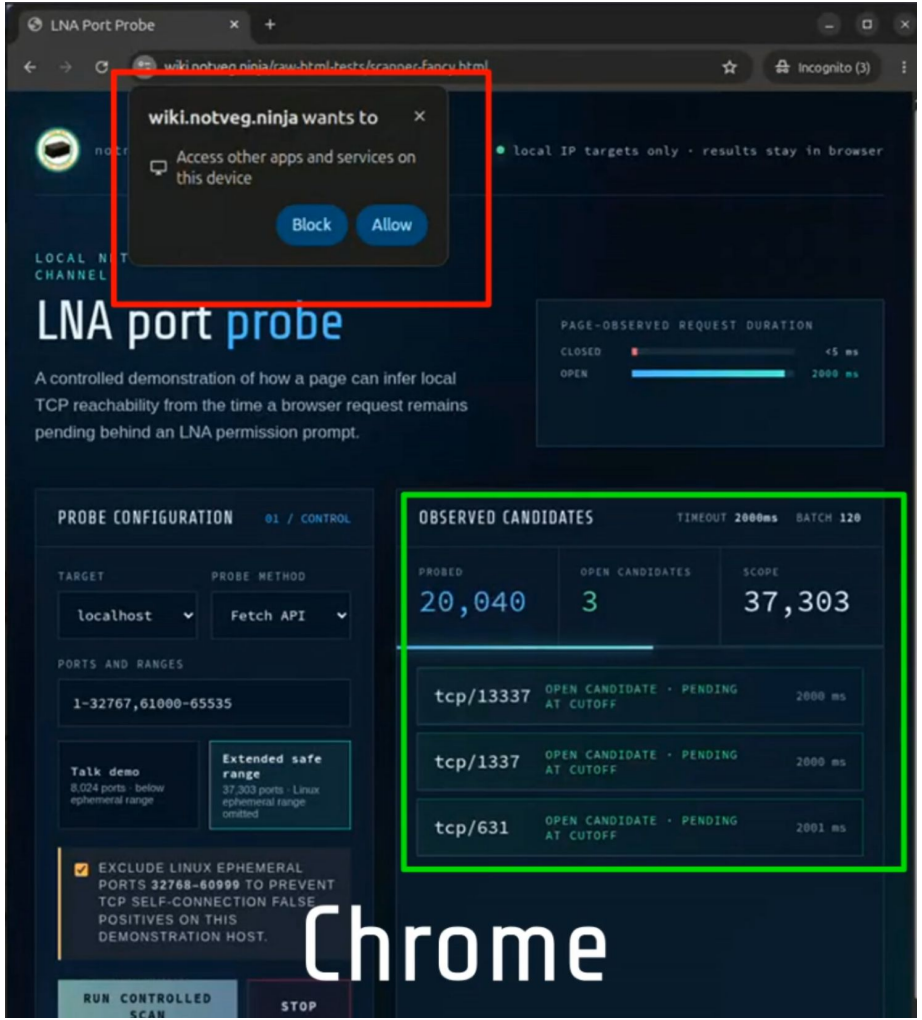
```
const c = new AbortController();                    // Controls request cancellation
setTimeout(() => c.abort(), 2000);                   // Abort after two seconds

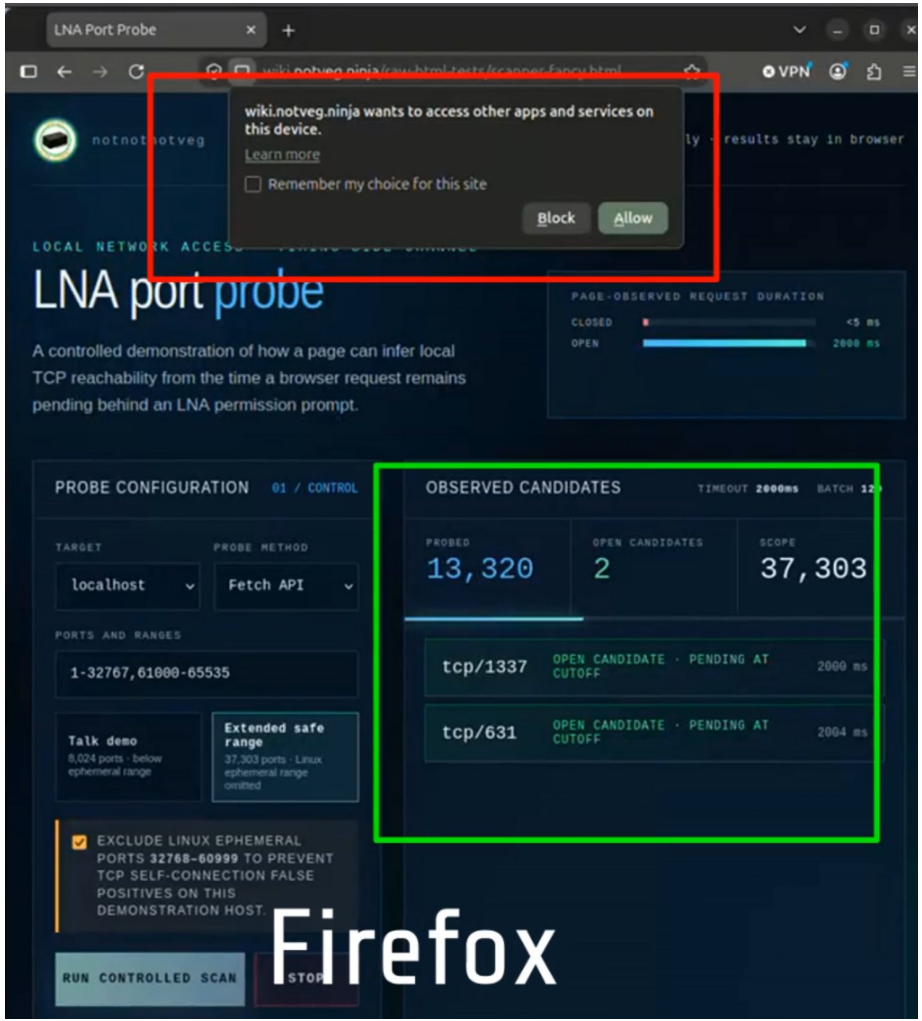
fetch("http://localhost:1337",                       // Start the local request
    { mode: "no-cors", signal: c.signal})      // Attach the cancellation signal
  .catch(e => console.log(
    e.name === "AbortError" ? "open" : "closed"   // Timeout -> request pending
));
```

Recorded PoC

```
gm@red-door:~$ nmap -Pn -p- localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-18 17:06 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000058s latency).
Not shown: 65532 closed tcp ports (conn-refused)
PORT      STATE SERVICE
631/tcp   open  ipp
1337/tcp  open  waste
13337/tcp open  unknown

Nmap done: 1 IP address (1 host up) scanned in 0.91 seconds
```


LNA Port Probe
wiki.notveg.ninja/raw-html-tests/scanner-fancy.html
Incognito (3)
wiki.notveg.ninja wants to
Access other apps and services on this device
Block
Allow
local IP targets only · results stay in browser
LNA port probe
A controlled demonstration of how a page can infer local TCP reachability from the time a browser request remains pending behind an LNA permission prompt.
PAGE-OBSERVED REQUEST DURATION
CLOSED
<5 ms
OPEN
2000 ms
PROBE CONFIGURATION
01 / CONTROL
TARGET
localhost
PROBE METHOD
Fetch API
PORTS AND RANGES
1-32767,61000-65535
Talk demo
8,024 ports · below ephemeral range
Extended safe range
37,303 ports · Linux ephemeral range omitted
EXCLUDE LINUX EPHEMERAL PORTS 32768-60999 TO PREVENT TCP SELF-CONNECTION FALSE POSITIVES ON THIS DEMONSTRATION HOST.
RUN CONTROLLED SCAN
STOP
OBSERVED CANDIDATES
TIMEOUT 2000ms
BATCH 120
PROBED
20,040
OPEN CANDIDATES
3
SCOPE
37,303
tcp/13337
OPEN CANDIDATE · PENDING AT CUTOFF
2000 ms
tcp/1337
OPEN CANDIDATE · PENDING AT CUTOFF
2000 ms
tcp/631
OPEN CANDIDATE · PENDING AT CUTOFF
2001 ms
Chrome


LNA Port Probe
wiki.notveg.ninja wants to access other apps and services on this device.
Learn more
Remember my choice for this site
Block
Allow
VPN
LOCAL NETWORK ACCESS
LNA port probe
A controlled demonstration of how a page can infer local TCP reachability from the time a browser request remains pending behind an LNA permission prompt.
PAGE-OBSERVED REQUEST DURATION
CLOSED
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EXCLUDE LINUX EPHEMERAL PORTS 32768–60999 TO PREVENT TCP SELF-CONNECTION FALSE POSITIVES ON THIS DEMONSTRATION HOST.
RUN CONTROLLED SCAN
STOP
OBSERVED CANDIDATES
TIMEOUT 2000ms
PROBED
13,320
OPEN CANDIDATES
2
SCOPE
37,303
tcp/1337
OPEN CANDIDATE · PENDING AT CUTOFF
2000 ms
tcp/631
OPEN CANDIDATE · PENDING AT CUTOFF
2004 ms
Firefox

Browser Support (Sept 2026)

BROWSER	SUPPORT	VERSION
Google Chrome	Supported	142+ (Oct 2025)
Chromium	Supported	143+ (Oct 2025)
Firefox	Supported (desktop)	153+ (Jul 2026)
Microsoft Edge	Supported	143+ (Dec 2025)
Safari / WebKit	In development	No release yet (Sept 2026)

Fixing the timing leak

LANDED
CHROME 144

PARTIAL MITIGATION

1. Known local URL
2. **Early prompt**
3. Request continues

IP literals · localhost / .localhost · .local

Prompt result is ignored; the authoritative LNA check still runs later.

STILL NEEDED

COMPLETE FIX

1. DNS resolve
2. Classify target
3. **Enforce permission**
4. **Open TCP**

The connection starts only after the authoritative check allows it.

Resolve first. Ask permission. Connect last.

Chrome Extension : LNAlyzer





In a nutshell

"May this webpage connect to the local service?"



BLOCK / ALLOW

"Is anything listening on this port?"

`.js`

FAST FAILURE / PENDING REQUEST

LNA gates the request to private networks.
JavaScript can still infer port state from request timing.

Thank you!

linkedin.com/in/gevmanekshaw

Resources · PoCs · References

https://github.com/notnotnotveg/bsides-nyc-2026-lna/wiki/Resources





API / SURFACE	CHROME	FIREFOX	NOTES
Fetch APIs	Yes	Yes	fetch(), XHR and EventSource
HTML resource loads	Yes	Yes	Images, scripts, styles and frames
Worker fetch()	Yes*	Yes*	Context and implementation dependent
WebSocket	147+	154+	Timing oracle not yet confirmed
WebTransport	147+	No	Firefox implementation pending
Beacon / hyperlink ping	Yes	Yes	No direct completion signal
WebRTC	No*	No	Outside the current LNA flow

* Worker behavior varies by context. WebRTC LNA is disabled by default in Chrome and unfinished in Firefox.